ISTA 322 - Data Engineering

College of Information Science

Fall 2025

Course Instructor:

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Course Description

This course will be inviting for a wide variety of students from across disciplines, and they will learn how to use industry standard tools and practices to make large data sets usable for scientists and other decision makers.
From data collection and preparation, to the creation of big data stores, databases, or systems to make data flow, this course will focus on the practical work needed to prepare big data for analyses across contexts. Students will be introduced to a variety of technical tools for data management, storage, use, and manipulation.

Course Objectives

The objective of this course is to train students in the tools and theory behind data engineering so that they can be deployed in the real world. This will revolve around extracting data from static and streaming sources, learning how to transform it in a way to be used for later analytics/machine learning, and data science applications, storing and querying that data in a

database. Additional emphasis will be placed on learning some cloud computing
Methods.

Expected Learning Outcomes

1. Students will be able to extract data from static and streaming data sources
2. Students will develop and apply skills for data cleaning, munging, and transformation so that they are usable for
later analysis.
3. Students will be able to generate and query SQL databases and access said databases from a Python interface
4. Students will understand major cloud and distributed computing technologies and be able to leverage them for data engineering applications.

Course Communications

We will use Piazza for course communication as well as Q/A. Please join the course Piazza page.

https://piazza.com/arizona/fall2025/ista322

Access Code: ofaqvq38tej

Required Readings

Learning Data Science, Sam Lau, Joey Gonzalez, and Deb Nolan, 2023.

https://learningds.org/intro.html

Grade Distribution

90-100%	A	exemplary, far beyond reqs/expectations
80-89%	B	exceeds requirements/expectations
70-79%	C	meets requirements/expectations
60-69%	D	falls short of requirements/expectations
< 60%	E	repeat of course needed

Required Materials

All students will need a functional laptop or personal computer.

We will use AWS services as well as OpenAI paid API, for which the students must use their credit card to pay for the used service. However, alternative approaches will be provided to students who don't have access to credit cards or do not want to pay for these learning experiences.

The guideline for effective use of these services will be provided however the students are completely responsible for their usage of these paid services.

Complete List of Assignments with Grade Breakdown

Activity	Total Percent	Activity & Notes
Programming assignments	35%	6 programming assignments
Final Project	20%	This is a full, independent data engineering project that will be completed in the final weeks of class.
Midterm Exam	15%	It is required
Final Exam	15%	It is required
Review Labs	15%	8-9 Labs

Grade dispute policy

If you believe there is an issue with your regrade, such as an incorrectly applied rubric or a miscalculation, you are required to provide a written and detailed explanation of the issue. The regrade request must be submitted to instructors (TA and me) via Gradescope (for homework or lab assignments) and Piazza (for exams).

Any **regrade request** must be submitted **within one week** after the release of the grades. After one week of release, your grades will be permanent and no change will be made.

Regrade request period for the **final exam** will be shorter due the grade submission deadline. The regrade request period for the final exam will be announced to the class by the release of the exam grades.

It is your responsibility to **double-check** your submission with respect to the submission instruction. **If you miss any deliverable you won't get any grade for that, and regrade request is not applicable.**

Late Submission Policy

1-24 hours late 10% penalty.

24-48 hours late 20% penalty.

No points after 48 hours.

If you have faced a special circumstance that you cannot submit on time you should get my **permission before the deadline** (sooner you ask more likely that I accept). **I won't extend any deadline after the submission due time. Submit your request via Piazza to instructors (TA and me) before the deadline along with your documentation!**

You won't be able to submit your work after 48 hours. We won't accept any submission by email or via Piazza.

Exam policy:

It is the student's responsibility to ensure uninterrupted internet access during the exam window. During the exam, you must record your entire desktop screen using Zoom, and the recording must clearly capture your face via a front-facing or external camera. The video recording must be submitted along with your exam. Failure to submit a recording will result in a grade of zero for the midterm. These recordings will be reviewed to investigate any potential violations of course policies, including the use of generative AI tools.

Midterm Exam Date:

The midterm exam will be open on Oct 17th at 8:00 am and will be closed on Oct 18th at 11:59 pm. It is the students responsibility to make sure that they have an uninterrupted internet access during the exam window.

Final Exam Date:

The final exam will be open on Dec 12th at 8:00 am and will be closed on Dec 13th at 11:59 pm. It is the students responsibility to make sure that they have an uninterrupted internet access during the exam window.

A high-level outline of topics:

- Introduction to Data Engineering
- Data Types and Working with Data
- Data Transformation
- Advanced Data Manipulation
- Data Collection and Web Scraping
- Data Transformation using Prompt Engineering and LLMs
- Relational Databases
- Bottom-Up Design and Database Normalization
- Data Warehousing
- NoSQL Databases
- Big Data Tools (pyspark)

Course Conduct and Campus Policies (be familiar with all campus policies)

1. Students are encouraged to share intellectual views and discuss freely the principles and applications of course materials. However, graded work/exercises must be the product of independent effort unless otherwise instructed. Students are expected to adhere to the UA Code of Academic Integrity as described in the UA General Catalog. See:http://deanofstudents.arizona.edu/academic-integrity/students/academic-integrity.

2. It is the University's goal that learning experiences be as accessible as possible. If you anticipate or experience physical or academic barriers based on disability or pregnancy, please let me know immediately so that we can discuss options. You are also welcome to contact Disability Resources (520-621-3268) to establish reasonable accommodations. For additional information on Disability Resources and reasonable accommodations, please visit http://drc.arizona.edu/.

3. The Arizona Board of Regents' Student Code of Conduct, ABOR Policy 5-308, prohibits threats of physical harm to any member of the University community, including to one's self. See: http://policy.arizona.edu/threatening- behavior-students.

4. All student records will be managed and held confidentially. http://www.registrar.arizona.edu/ferpa/default.htm

5. Information contained in the course syllabus, other than the grade and absence policy, may be subject to change

   with advance notice, as deemed appropriate by the instructor.

6. UA Non-discrimination and Anti-harassment policy: http://policy.arizona.edu/human-resources/nondiscrimination-

   and-anti-harassment-policy.

7. Confidentiality of Student Records: http://www.registrar.arizona.edu/ferpa/default.htm.

8. Information contained in this syllabus, other than the grade and absence policy, may be subject to change without advance notice as deemed appropriate by the instructor.

Generative AI

While the use of generative AI tools such as ChatGPT for the purpose of learning and practicing is allowed, the students are not allowed to submit the output of AI tools as their own work. The use of AI tools such as ChatGPT or smart coding assistant during the exam is prohibited and considered as plagiarism.

Additional Resources for Students

UA Academic policies and procedures are available at http://catalog.arizona.edu/policies

Campus Health

http://www.health.arizona.edu/

Campus Health provides quality medical and mental health care services through virtual and in-person care.
Phone: 520-621-9202

Counseling and Psych Services (CAPS)
https://health.arizona.edu/counseling-psych-services
CAPS provides mental health care, including short-term counseling services.
Phone: 520-621-3334

The Dean of Students Office's Student Assistance Program
http://deanofstudents.arizona.edu/student-assistance/students/student-assistance
Student Assistance helps students manage crises, life traumas, and other barriers that impede success. The staff addresses the needs of students who experience issues related to social adjustment, academic challenges, psychological health, physical health, victimization, and relationship issues, through a variety of interventions, referrals, and follow up services.
Email: DOS-deanofstudents@email.arizona.edu
Phone: 520-621-7057

Survivor Advocacy Program
https://survivoradvocacy.arizona.edu/
The Survivor Advocacy Program provides confidential support and advocacy services to student survivors of sexual and gender-based violence. The Program can also advise students about relevant non-UA resources available within the local community for support.
Email: survivoradvocacy@email.arizona.edu
Phone: 520-621-5767

University-wide Policies link

Links to the following UA policies are provided here:

http://catalog.arizona.edu/syllabus-policies:

- Absence and Class Participation Policies
- Threatening Behavior Policy
- Accessibility and Accommodations Policy
- Code of Academic Integrity
- Nondiscrimination and Anti-Harassment Policy
- Subject to Change Statement